Roll no : 241501199
Name : Sherwin Richie J

CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -1

AIM: - Study of various Network commands used in Linux and Windows:

ALGORITHM / PROCEDURE:

1. Open the Terminal in Linux and the Command Prompt (cmd) in Windows.
2. Use hostnameto see the name of the computer.
3. Use ifconfig(Linux) and ipconfig(Windows) to see the IP address, subnet mask and MAC address.
4. Use pingto check whether the remote system is reachable. Note the packet loss and the time taken.
5. Use traceroute(Linux) and tracert(Windows) to see the path taken by the packets.
6. Use nslookupand digto find the IP address of a domain name.
7. Use netstatto see the active connections and open ports.
8. Use arp -ato see the IP address to MAC address table.
9. Use route(Linux) and route print(Windows) to see the routing table.
10. Note down the output of each command.

COMMANDS USED:

Purpose	Linux	Windows
Host name	`hostname`	`hostname`
IP configuration	`ifconfig / ip addr`	`ipconfig /all`
Check connectivity	`ping -c 4 google.com`	`ping google.com`
Trace the route	`traceroute google.com`	`tracert google.com`
DNS lookup	`nslookup / dig`	`nslookup`
Active connections	`netstat -tuln`	`netstat -an`
ARP table	`arp -a`	`arp -a`
Routing table	`route -n`	`route print`
MAC address	`ip link`	`getmac`

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OUTCOMES:

D:\>hostname

LAPTOP-O0K6G7B7

D:\>ipconfig/all

Windows IP Configuration

Host Name . . . . . : LAPTOP-O0K6G7B7

Primary Dns Suffix . . . . . :

Node Type . . . . . : Mixed

IP Routing Enabled. . . . . : No

WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 4:

Connection-specific DNS Suffix . :

Description . . . . . : VirtualBox Host-Only Ethernet Adapter

Physical Address. . . . . : 0A-00-27-00-00-02

DHCP Enabled. . . . . : No

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::1da4:d7dd:72a8:cf41%2(Preferred)

IPv4 Address. . . . . : 192.168.56.1(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . . . :

DHCPv6 IAID . . . . . : 352976935

DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-C9-4C-BB-90-E8-68-14-92-7B

NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 9:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. . . . . : 92-E8-68-14-92-2B

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. . . . . : 92-E8-68-14-92-3B

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :

Description . . . . . : VMware Virtual Ethernet Adapter for VMnet1

Physical Address. . . . . : 00-50-56-C0-00-01

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::86fd:79b9:4558:7496%5(Preferred)

IPv4 Address. . . . . : 192.168.147.1(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Lease Obtained. . . . . : 31 August 2026 09:52:47

Lease Expires . . . . . : 31 August 2026 10:22:48

Default Gateway . . . . . :

DHCP Server . . . . . : 192.168.147.254

DHCPv6 IAID . . . . . : 1023430742

DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-C9-4C-BB-90-E8-68-14-92-7B

NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :

Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8

Physical Address. . . . . : 00-50-56-C0-00-08

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::c2e3:901d:6ad1:4bcb%13(Preferred)

IPv4 Address. . . . . : 192.168.75.1(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Lease Obtained. . . . . : 31 August 2026 09:52:52

Lease Expires . . . . . : 31 August 2026 10:22:48

Default Gateway . . . . . :

DHCP Server . . . . . : 192.168.75.254

DHCPv6 IAID . . . . . : 1056985174

DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-C9-4C-BB-90-E8-68-14-92-7B

Primary WINS Server . . . . . : 192.168.75.2

NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

Description . . . . . : MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter

Physical Address. . . . . : 90-E8-68-14-92-7B

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::ed17:b200:1378:289f%15(Preferred)

IPv4 Address. . . . . : 172.30.4.22(Preferred)

Subnet Mask . . . . . : 255.255.224.0

Lease Obtained. . . . . : 31 August 2026 08:45:45

Lease Expires . . . . . : 01 September 2026 08:45:44

Default Gateway . . . . . : 172.30.0.1

DHCP Server . . . . . : 172.30.0.1

DHCPv6 IAID . . . . . : 227600488

DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-C9-4C-BB-90-E8-68-14-92-7B

DNS Servers . . . . . : 8.8.8.8

NetBIOS over Tcpip. . . . . : Enabled

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Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Bluetooth Device (Personal Area Network)

Physical Address. . . . . : 90-E8-68-14-92-7A

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

D:\>arp -a

Interface: 192.168.56.1 --- 0x2

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.147.1 --- 0x5

Internet Address	Physical Address	Type
192.168.147.254	00-50-56-e9-44-86	dynamic
192.168.147.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static

224.0.0.251 01-00-5e-00-00-fb static

224.0.0.252 01-00-5e-00-00-fc static

239.255.255.250 01-00-5e-7f-ff-fa static

255.255.255.255 ff-ff-ff-ff-ff-ff static

Interface: 192.168.75.1 --- 0xd

Internet Address Physical Address Type

192.168.75.254 00-50-56-e6-63-f5 dynamic

192.168.75.255 ff-ff-ff-ff-ff-ff static

224.0.0.22 01-00-5e-00-00-16 static

224.0.0.251 01-00-5e-00-00-fb static

224.0.0.252 01-00-5e-00-00-fc static

239.255.255.250 01-00-5e-7f-ff-fa static

255.255.255.255 ff-ff-ff-ff-ff-ff static

Interface: 172.30.4.22 --- 0xf

Internet Address Physical Address Type

172.30.0.1 cc-82-7f-30-b5-a3 dynamic

172.30.31.255 ff-ff-ff-ff-ff-ff static

224.0.0.22 01-00-5e-00-00-16 static

224.0.0.251 01-00-5e-00-00-fb static

224.0.0.252 01-00-5e-00-00-fc static

239.255.102.18 01-00-5e-7f-66-12 static

239.255.255.250 01-00-5e-7f-ff-fa static

```
  255.255.255.255        ff-ff-ff-ff-ff-ff     static


D:\>netstat -r

Interface List

  2...0a 00 27 00 00 02 ......VirtualBox Host-Only Ethernet Adapter

  4...92 e8 68 14 92 2b ......Microsoft Wi-Fi Direct Virtual Adapter

 17...92 e8 68 14 92 3b ......Microsoft Wi-Fi Direct Virtual Adapter #2

  5...00 50 56 c0 00 01 ......VMware Virtual Ethernet Adapter for VMnet1

 13...00 50 56 c0 00 08 ......VMware Virtual Ethernet Adapter for VMnet8

 15...90 e8 68 14 92 7b ......MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter

 18...90 e8 68 14 92 7a ......Bluetooth Device (Personal Area Network)

  1...........................Software Loopback Interface 1

IPv4 Route Table

Active Routes:

Network Destination        Netmask          Gateway       Interface  Metric

          0.0.0.0          0.0.0.0       172.30.0.1      172.30.4.22     35

        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331

        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331

  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331

       172.30.0.0    255.255.224.0         On-link       172.30.4.22    291

      172.30.4.22  255.255.255.255         On-link       172.30.4.22    291

    172.30.31.255  255.255.255.255         On-link       172.30.4.22    291
```

192.168.56.0	255.255.255.0	On-link	192.168.56.1	281
192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
192.168.75.0	255.255.255.0	On-link	192.168.75.1	291
192.168.75.1	255.255.255.255	On-link	192.168.75.1	291
192.168.75.255	255.255.255.255	On-link	192.168.75.1	291
192.168.147.0	255.255.255.0	On-link	192.168.147.1	291
192.168.147.1	255.255.255.255	On-link	192.168.147.1	291
192.168.147.255	255.255.255.255	On-link	192.168.147.1	291
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.147.1	291
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	192.168.75.1	291
224.0.0.0	240.0.0.0	On-link	172.30.4.22	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.147.1	291
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	On-link	192.168.75.1	291
255.255.255.255	255.255.255.255	On-link	172.30.4.22	291

Persistent Routes:

None

IPv6 Route Table

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Active Routes:

If	Metric	Network Destination	Gateway
1	331	::1/128	On-link
5	291	fe80::/64	On-link
2	281	fe80::/64	On-link
13	291	fe80::/64	On-link
15	291	fe80::/64	On-link
2	281	fe80::1da4:d7dd:72a8:cf41/128	On-link
5	291	fe80::86fd:79b9:4558:7496/128	On-link
13	291	fe80::c2e3:901d:6ad1:4bcb/128	On-link
15	291	fe80::ed17:b200:1378:289f/128	On-link
1	331	ff00::/8	On-link
5	291	ff00::/8	On-link
2	281	ff00::/8	On-link
13	291	ff00::/8	On-link
15	291	ff00::/8	On-link

Persistent Routes:

None

Roll no : 241501199
Name : Sherwin Richie J

D:\>nslookup www.google.com

Server: dns.google

Address: 8.8.8.8

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:4828:7700::

2001:4860:4827:7700::

2001:4860:482b:7700::

2001:4860:482d:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

2001:4860:4826:7700::

2001:4860:4829:7700::

142.251.153.119

142.251.156.119

142.251.154.119

142.251.150.119

142.251.157.119

142.251.152.119

142.251.151.119

142.251.155.119

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Name : Sherwin Richie J

D:\>ping google.com

Pinging google.com [142.250.66.14] with 32 bytes of data:

Reply from 142.250.66.14: bytes=32 time=5ms TTL=118

Reply from 142.250.66.14: bytes=32 time=6ms TTL=118

Reply from 142.250.66.14: bytes=32 time=23ms TTL=118

Reply from 142.250.66.14: bytes=32 time=26ms TTL=118

Ping statistics for 142.250.66.14:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 5ms, Maximum = 26ms, Average = 15ms

D:\>route print

Interface List

2...0a 00 27 00 00 02 ......VirtualBox Host-Only Ethernet Adapter

4...92 e8 68 14 92 2b ......Microsoft Wi-Fi Direct Virtual Adapter

17...92 e8 68 14 92 3b ......Microsoft Wi-Fi Direct Virtual Adapter #2

5...00 50 56 c0 00 01 ......VMware Virtual Ethernet Adapter for VMnet1

13...00 50 56 c0 00 08 ......VMware Virtual Ethernet Adapter for VMnet8

15...90 e8 68 14 92 7b ......MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter

18...90 e8 68 14 92 7a ......Bluetooth Device (Personal Area Network)

1...........................Software Loopback Interface 1

Roll no : 241501199
Name : Sherwin Richie J

IPv4 Route Table

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	172.30.0.1	172.30.4.22	35
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
172.30.0.0	255.255.224.0	On-link	172.30.4.22	291
172.30.4.22	255.255.255.255	On-link	172.30.4.22	291
172.30.31.255	255.255.255.255	On-link	172.30.4.22	291
192.168.56.0	255.255.255.0	On-link	192.168.56.1	281
192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
192.168.75.0	255.255.255.0	On-link	192.168.75.1	291
192.168.75.1	255.255.255.255	On-link	192.168.75.1	291
192.168.75.255	255.255.255.255	On-link	192.168.75.1	291
192.168.147.0	255.255.255.0	On-link	192.168.147.1	291
192.168.147.1	255.255.255.255	On-link	192.168.147.1	291
192.168.147.255	255.255.255.255	On-link	192.168.147.1	291
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.147.1	291
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281

224.0.0.0 240.0.0.0 On-link 192.168.75.1 291

224.0.0.0 240.0.0.0 On-link 172.30.4.22 291

255.255.255.255 255.255.255.255 On-link 127.0.0.1 331

255.255.255.255 255.255.255.255 On-link 192.168.147.1 291

255.255.255.255 255.255.255.255 On-link 192.168.56.1 281

255.255.255.255 255.255.255.255 On-link 192.168.75.1 291

255.255.255.255 255.255.255.255 On-link 172.30.4.22 291

Persistent Routes:

None

IPv6 Route Table

Active Routes:

If Metric Network Destination Gateway

1 331 ::1/128 On-link

5 291 fe80::/64 On-link

2 281 fe80::/64 On-link

13 291 fe80::/64 On-link

15 291 fe80::/64 On-link

2 281 fe80::1da4:d7dd:72a8:cf41/128

On-link

5 291 fe80::86fd:79b9:4558:7496/128

On-link

13 291 fe80::c2e3:901d:6ad1:4bcb/128

On-link

15 291 fe80::ed17:b200:1378:289f/128

On-link

1 331 ff00::/8 On-link

5 291 ff00::/8 On-link

2 281 ff00::/8 On-link

13 291 ff00::/8 On-link

15 291 ff00::/8 On-link

Persistent Routes:

None

D:\>tracert google.com

Tracing route to google.com [142.250.66.14]

over a maximum of 30 hops:

1 26 ms 1 ms 1 ms 172.30.0.1

2 19 ms 2 ms 3 ms 137.97.163.129

3 * * * Request timed out.

4 10 ms 8 ms 5 ms 172.16.12.64

5 9 ms 6 ms 6 ms 172.16.12.64

6 44 ms 6 ms 5 ms 72.14.215.136

7 6 ms 9 ms 6 ms 142.251.70.199

8 48 ms 6 ms 6 ms 142.251.60.187

9 24 ms 6 ms 6 ms bom07s35-in-f14.1e100.net [142.250.66.14]

Trace complete.

Linux commands :

traceroute google.com

$ traceroute google.com

traceroute to google.com (142.250.183.14), 30 hops max, 60 byte packets

1 192.168.1.1 2.123 ms 1.987 ms 2.015 ms

2 10.20.0.1 8.432 ms 8.210 ms 8.345 ms

3 100.64.0.1 15.234 ms 15.102 ms 15.187 ms

4 142.250.183.14 25.456 ms 25.321 ms 25.402 ms

ip addr

$ ip addr

1: lo: <LOOPBACK,UP,LOWER_UP>

inet 127.0.0.1/8 scope host lo

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP>

inet 192.168.1.10/24 brd 192.168.1.255

inet6 fe80::1234:5678:abcd:ef01/64

netstat -ano

$ netstat -ano

Active Internet connections

Proto Local Address Foreign Address State

tcp 0.0.0.0:22 0.0.0.0:* LISTEN

tcp 192.168.1.10:45678 142.250.183.14:443 ESTABLISHED

tcp 192.168.1.10:52341 142.250.72.14:443 ESTABLISHED

ip route

$ ip route

default via 192.168.1.1 dev eth0

192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.10

RESULT:

Thus the various network commands used in Linux and Windows were studied and executed successfully